Signaled Bandits

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Abstract

This paper introduces the signaled bandit game, a generalization of experts' problems and multi-armed bandit problems in online learning. In both settings, the learner tries to maximize the cumulative sum of rewards by playing one arm at a time. The difference between the two lies in observability: in experts' problems, the learner observes the rewards of all arms regardless of her arm selection, while in bandit problems, she only observes the reward of the arm she plays. The signaled bandit game interpolates across these two regimes through the notion of signal informativeness, such that the learner accrues the reward of the selected arm but also observes a noisy signal for all the unplayed arms. As signals become more informative, the incentives to explore decrease and the game converges to the experts' setting. Conversely, as signals become uninformative, intentional exploration becomes necessary and the game collapses to a bandit setting. I first present a smooth worst-case lower bound and a discontinuous instance-dependent lower bound which features an abrupt transition from finite to asymptotic regret as signals become less informative. I then show that the optimal algorithms in the limiting regimes have little success in the generalized game: pure-exploitation algorithms suffer linear regret in the presence of uninformative signals, and `UCB` algorithms cannot achieve finite regret even with very informative signals. Instead, I propose a simple variation of `UCB` which combines the best of both worlds: it preserves sub-logarithmic regret uniformly and achieves finite regret for mildly informative signals. Finally, I argue that signaled bandits can be used to model exploration-exploitation settings in the presence of public low-quality information, including applications to portfolio selection, market entry, treatment effects and fraud detection.

1 Introduction

There exists two canonical problems in online learning: the experts' problem and the bandit problem, both with a myriad of applications to finance, marketing, clinical research, experimental design in economics and others.[1] In the stochastic version of both games, the learner selects one arm in every period and obtains a noisy reward from that arm's distribution with unknown expectation. The goal of the learner is to maximize the expected cumulative sum of rewards. The main difference across settings is that, in experts' problems, the learner gets to observe the outcome of counterfactual arms.

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[1] Cesa-Bianchi and Lugosi (2006); Bubeck and Cesa-Bianchi (2012); Cover (1991); Schwartz et al. (2017); Villar et al. (2015); Kasy and Sautmann (2021).

By contrast, in bandit problems, the learner only observes the reward of the selected arm. A simple instance of the experts' problem is the *portfolio selection* problem (Cover, 1991). Every period the learner tries to conform the best possible portfolio by mixing a set of publicly traded stocks with unknown expectation. At the end of the period, she collects the returns of her portfolio and observes the returns of all assets, so she can compute the returns of every counterfactual portfolio. Similarly, a standard example of the bandit problem is the *AB testing* problem in marketing. Every period, the learner decides which add to show to a customer. She then observes the behavior of the customer conditional on the add, but she can never observe the customer's behavior for the counterfactual add.

One may imagine an intermediate regime where the learner observes the reward of the selected arm and a *flow* of noisy signals for all arms independent of her selection. More concretely, I assume that signals are weakly less informative about the arm's expectation than rewards. This notion of signal informativeness allows for a nice interpolation across regimes. When signals are as informative as rewards, the problem reduces to the experts' game. Conversely, as signals become less informative, the problem collapses to a bandit game. I refer to the general bandit problem with side signals as the **signaled bandit** game. This game can be used to model traditional exploration-exploitation settings where the learner has access to a free flow of low quality information, including applications to Finance and Economics.

Example 1 - Portfolio Selection and Public Data. Every quarter an investor selects a portfolio of non-publicly listed companies, and collects the returns of her investment at the end of the period. Although returns for counterfactual companies are not openly disclosed, the learner can get a rough idea of their returns by looking at public company reports.

Example 2 - Market Entry and Data Frequency. Every period, a firm observes some low-frequency data on demand for different markets—for instance, coming from public information releases—and decides which one it will enter for that period. Beyond capturing demand in the selected market, market entry has an additional information value to the company as it gets access to high-frequency data on demand.

Example 3 - Treatment Allocation with Scientific Communication. Consider an experimental setting where a policymaker is trying to maximize the cumulative welfare of her population by applying one in several treatments with unknown treatment effects. For instance, governments were making active decisions on vaccine suppliers or lockdown strategies during Covid-19. Assume that the ATT of each treatment is constant across populations. When selecting a treatment for her population, the policymaker can restore on publicly released health information coming from other countries. Such information is weakly less informative than internal information.

Example 4 - Fraud Detection and Effort. Many public and private agencies reward their workers by the amount of detected fraud, including tax evasion, software breaches, legal misconduct, contracts violation, etc. Every period the fraud officer inspects a case of type $k \in [K]$. For instance, cases might be split across industry, ownership, region, etc. If there is a fraud violation x of scale x_1 or more, the inspector will flag it and she will receive a bounty as compensation. All unselected cases are parsed by a machine that only detects salient cases of fraud, i.e. it flags them if the fraud scale is $x > x_2 > x_1$. The inspector has access to the history of cases parsed by the machine, as well as her own history. The

machine information can be modeled as a flow of low quality information compared to the inspector's decisions.

Example 5 - Treatment Allocation with Forced Exploration. Often times, the scope of experimentation is limited by regulatory or logistic considerations. For instance, a transportation company might be forced to offer a baseline service for all K routes, but she gets extra capacity to operate additional services on these routes at a profit. Similarly, a researcher might decide how to split her time between teaching, administrative duties and research, but she must commit at least two hours in her day to each activity. Because neither the transportation company nor the researcher optimize over the forced exploration segment, and assuming that the more time the learner spends on a route or task the more information she obtains on the expected return of that arm, forced exploration can be modeled as a flow of low quality information

Incentives for Exploration. Intuitively, as side-signals become more informative, the learner has a reduced incentive for dedicated exploration and can simply focus on reward maximization. This pattern is prevalent in all five examples above. For example, in the portfolio selection problem, as the quality of firm's public data increases, the learner perceives a reduced incentive to hold assets only for information purposes. Similarly, in the case of fraud detection, as $x_2 \to x_1$, the inspector has a reduced incentive to investigate cases different from those statistically flagged by the machine. Optimal algorithms in the limiting regimes also reflect this dynamic. In the experts' problem, pure-exploitation algorithms—like Follow-the-Leader (`FtL`)—turn out to be optimal. However, they suffer from linear regret in bandit problems. Conversely, algorithms with dedicated exploration—like Upper Confidence Bound (`UCB`)—turn out to be optimal in bandit problems but they remain overly conservative in games with full-information. This pattern suggests that an optimal algorithm in the general signaled bandit game should feature automatic exploration reduction as signals become increasingly informative.

Asymptotic vs Finite Regret. The main challenge of this interpolation exercise is the qualitatively different nature of instance-dependent regret across regimes. While finite regret is possible in the experts' problem, no algorithm can achieve sub-logarithmic regret in the bandit problem. In other words, there is a finite-to-asymptotic regret gap between these two regimes. This abrupt transition poses the existence of a tipping point along the dimension of signal informativeness below which no algorithm can obtain finite regret, and above which finite regret becomes possible. This gap also manifests in the actual derivation of lower bounds, as standard procedures deliver empty bounds in the presence of a *linear* source of information (Degenne et al., 2018). This forces the analyst to rely on unconventional tools to characterize the difficulty of the problem along the signal informativeness dimension as in Garivier et al. (2019). On the other hand, worst-case regret is more amenable to an interpolation analysis as both regimes display nearly-tight lower bounds on regret of $O(\sqrt{N})$.

Contributions. I first derive novel worst-case lower bounds for the general signaled bandit game of $O(\sqrt{N})$ which interpolate nicely between standard lower bounds in the experts' and bandit problems. I leverage this result to argue that the experts' problem is indeed the easiest instance in a signaled bandit game and the bandit problem is the hardest instance. I then characterize instance dependent lower bounds, with a particular emphasis on the finite to asymptotic regret transition point. As a glimpse, if N signals are weakly less informative than a single reward, then, the game is no different from a bandit setting and no algorithm could be expected to achieve finite regret.

In terms of algorithmic contributions, I first show that pure-exploitation algorithms cannot achieve sub-linear regret in the general signaled bandit. In particular, the regret of pure exploitation algorithms is of the same order than the informational content of signals. As a result, if the learner needs $\ln(N)$ signals to obtain the informational equivalent of a single reward, then pure-exploitation algorithms will incur $O(\ln(N))$ regret. I then show that Signaled UCB (`SUCB`)—a small variation of a `UCB` algorithm which uses a precision-weighted average and refined bounds—combines many desirable properties. First, it displays automatic exploration reduction. As signals become more informative, effective bounds converge to 0, so the algorithm behaves as a pure-exploitation policy. By contrast, when signals become entirely uninformative, bounds converge to standard `UCB` bounds. Second, I show that this algorithm achieves finite regret for mildly informative signals and achieves sublinear regret uniformly in the signaled bandit game. In fact, the transition from asymptotic to finite regret happens at a signal informational content of order $N\Delta$, just like the one identified in the lower-bounds, what suggest that my lower bounds are order-tight and that `SUCB` is instance-dependent optimal and minmax near-optimal in the signaled bandit game.

My results reveal that remarkably low quality information of unplayed arms can significantly reduce the need for dedicated exploration, improving substantially the speed of learning. As such, the analyst should not neglect free information. However, completely disregarding exploration can also be problematic, especially with borderline informative signals. `SUCB` provides robust guarantees in the entire spectrum of signaled bandits with important applications to settings with free low-quality information.

The rest of this manuscript is structured as follows. Section 2 introduces the signaled bandit game and discusses the experts' problem and the bandit problem as especial cases. Section 3 derives smooth worst-case lower bounds and uses them to argue that the experts' problem is the easiest signaled bandit instance, and that the bandit problem is the hardest signaled bandit instance. I further characterize instance-dependent lower bounds and the transition point from finite to asymptotic regret. Section 4 presents an unbounded lower bound for pure-exploitation algorithms in the signaled bandit game, introduces the main algorithm of the manuscript, and shows some of its properties including automatic exploration reduction, instance-dependent optimality and near minmax optimality. Section 5 confirms the theoretical predictions through a large array of simulations. Finally, Section 6 concludes. All proofs can be found in Appendices A to C.

2 Model

Information Design. Consider a sequential decision problem over N periods indexed by i. Define the environment $\nu = (\nu_k)_k = (\nu_{kk'})_{kk'}$ as a collection of $K \times K$ signal distributions where $\nu_{kk'}$ is a $\sigma_{kk'}$-subgaussian random variable with $\mathbb{E}[\nu_{kk'}] = \mu_{k'}$ for all k'.[2] The sub-gaussian constant σ will be the main notion of informativeness in this manuscript.[3] Without loss of generality assume that $\mu_1 > \mu_2 \geq \cdots \geq \mu_K$, such that $\Delta_k = \mu_1 - \mu_k > 0$ where $\Delta_k \in (0, 1]$ for all k. For simplicity I restrict the environment to the symmetric variance class, i.e.

$$\sigma^2_{kk'} = \sigma^2_b(N) \;\; \forall k \neq k', \qquad \sigma^2_{kk} = \sigma^2_a < \infty, \qquad 0 < \sigma^2_a \leq \sigma^2_b(N) \leq \infty$$

[2] A random variable is σ-subgaussian if and only $\mathbb{E}[\exp(\lambda X)] \leq \exp(\lambda^2\sigma^2/2)$ for all $\lambda \in \mathbb{R}_+$.

[3] A completely parallel analysis would go through with sub-exponential constants or domain restrictions. Although I assume that σ is known to the learner, an upper bound on σ suffices for the purpose of this analyses at the expense of some conservativeness.

The notation implies that signal informativeness $\sigma_b^2(N)$ is allowed to vary with N. Without this restriction, pure exploitation will always remain order optimal. By allowing $\sigma_b^2(N)$ to vary with N, an assumption similar to those in local-to-zero asymptotics and the weak-IV literature, I can better represent frameworks with poorly informative signals, yet not entirely uninformative. When $\sigma_b^2(N) = \infty$ for fixed N, I assume that the learner observes a completely uninformative signal (or no signal at all) for all unplayed arms.[4] Every period, the learner gets to play an arm vector $k_i \in [K]$. When playing arm k in period i, she observes an *iid* stochastic realization of the signal vector $\nu_{k_i} = (\nu_{k_i,1}, \nu_{k_i,2}, \ldots, \nu_{k_i,K})$ and accrues a reward $\nu_{k_i,k_i,i}$.

Definition 2.1. *Signaled Bandit Environment Class. Let $\mathcal{E}$ be the collection of environments ν with*

1. *finite mean: $\mathbb{E}[\nu_{kk}] = \mu_k < \infty$, and bounded gaps $\Delta_k \in (0, 1]$ for all k,*

2. *σ_a-subgaussian rewards and $\sigma_b(N)$-subgaussian signals,*

3. *finite reward informativeness $\sigma_a^2 < \infty$ and weakly less informative signals $\sigma_a^2 \leq \sigma_b^2(N) \leq \infty$.*

Definition 2.2. *Δ Sub-Environment Class. For any environment class $\mathcal{E}$, I write $\mathcal{E}(\Delta) \subset \mathcal{E}$ for the Δ instance-dependent class where $\Delta_k \geq \Delta > 0$ for all $k \neq 1$.*

Interpretation of Information Design. Signal informativeness is the interpolation mechanism in this generalization of online learning games. I find this mechanism more empirically grounded than alternative interpolation devices where the authors assume that, occasionally, the learner may observe signals with the same information content as rewards. For instance, Degenne et al. (2018) assume that with probability p the learner observes the reward of an unplayed arm, and Mannor and Shamir (2011) assume that feedback structures can be represented through directed graphs, such that playing an arm reveals the rewards of all the arms connected to the selected arm in the graph. Additionally, I find the notion of signal informativeness more theoretically grounded than its competitors, as this concept presents a long tradition in the fields of microeconomic theory and information economics going back to Blackwell (1953) and Marschak and Miyasawa (1968).

The Problem of the Learner. Define the history $H_i = ((k_1, \nu_{k_1,1}), (k_2, \nu_{k_2,2}), \ldots, (k_{i-1}, \nu_{k_{i-1},i-1}))$ as the collection of selected arms and signal realizations up to period i. Observe that signals and rewards remain *iid* conditional on the history. Define a policy $\pi : H_i \to [K]$ as an adapted mapping from the history to the arm space $[K] = \{1, \ldots, k, \ldots, K\}$. Let k_i^π be the arm recommended by policy π in period i. The goal of the learner is to maximize in expectation the worst-case cumulative sum of rewards over some environment class $\mathcal{E}$, i.e.

$$\max_{\pi} \min_{\nu \in \mathcal{E}} \mathbb{E}_{\nu,\pi} \left[\sum_{i=1}^{N} \nu_{k_i^\pi, k_i^\pi, i} \right]$$

where the expectation is wrt to the (filtration generated by the) interaction between the history and policy. As it is customary in the learning literature, I look at the equivalent problem of regret

[4] I will show that one needs not to set $\sigma_b^2(N) = \infty$ to represent completely uninformative signals. In particular, signals of order $\Theta(N)$ are equivalently uninformative to the learner.

minimization. Let $\mu^*(\nu) = \max_k \mu_k$, then

$$\min_{\pi} \max_{\nu \in \mathcal{E}} \mathcal{R}(\pi, \nu, N) = \min_{\pi} \max_{\nu \in \mathcal{E}} N\mu^*(\nu) - \mathbb{E}_{\nu,\pi}\left[\sum_{i=1}^{N} \nu_{k_i^\pi, k_i^\pi, i}\right] \geq 0 \tag{1}$$

where $\mathcal{R}(\pi, \nu, N)$ is the regret of policy π in environment ν over N periods. A policy π is said to have regret of order $O(N^\alpha)$ in the environment class $\mathcal{E}$ iff $\max_{\nu \in \mathcal{E}} \mathcal{R}(\pi, \nu, N) \leq C_1 \cdot N^\alpha$, for some constant $C_1 < \infty$ and $\alpha \in [0, 1]$. Similarly, an environment class $\mathcal{E}$ is said to have a lower bound of order $\Omega(N^\beta)$ if $\max_{\nu \in \mathcal{E}} \mathcal{R}(\pi, \nu, N) \geq C_2 \cdot N^\beta$, for some constant C_2. Finally, a policy is said to be optimal in a class if its worst-case regret within that class matches the lower bound of the problem up to constant terms, i.e. $\alpha = \beta$. A policy is said to be near-optimal in a class if its worst-case regret within that class matches the lower bound of the problem up to logarithmic terms, i.e. $C_1 \cdot N^\alpha / (C_2 \cdot N^\beta) \leq C_3 \ln N$ for some finite constant C_3.

Limiting and Intermediate Regimes. The signaled bandit environment is a generalization of experts' and bandit regimes.

a) **Experts' Problems:** $\mathcal{E}_{\text{exp}} = \{\nu \in \mathcal{E} : \sigma_a^2 = \sigma_b^2(N), \ \forall N\}$. Signals are as informative as rewards, therefore there is no trade-off between learning (exploration) and reward maximization (exploitation).

b) **Bandit Problems:** $\mathcal{E}_{\text{bandit}} = \{\nu \in \mathcal{E} : \sigma_a^2 < M < \sigma_b^2(N) = \infty, \ \forall N\}$. Signals are completely uninformative while rewards remain informative. Information acquisition is tightly linked to rewards, therefore a critical trade-off emerges between learning and reward maximization.

c) **Additional Definitions:** $\mathcal{E}_{\text{exp}}(\Delta)$ and $\mathcal{E}_{\text{bandit}}(\Delta)$ are defined as expected. $\mathcal{E}_{\overline{\text{exp}}} = \mathcal{E} \setminus \mathcal{E}_{\text{exp}}$.

3 Lower Bounds in the Signaled Bandit Problem

Interpolating Lower Bounds. The signaled-bandit framework is smooth by construction. As the informational content of signals decreases, one may expect a smooth transition from the experts' regime to the bandit regime. In fact, one may wonder if this conceptual smoothness can be mapped into a smooth lower bound which interpolates between known results in the experts' and bandit games. The answer to this question is positive for worst-case (or minmax) lower bounds as both experts' and bandit lower bounds are of order $O(\sqrt{N})$, what eases the interpolation exercise. The task is more challenging for instance-dependent bounds, where experts' games are known to have tight finite regret bounds, and bandit games are known to have tight logarithmic bounds. In other words, there is qualitative gap between finite and asymptotic regret that hinders the smooth interpolation exercise. Yet, I leverage the smooth nature of the signaled bandit framework to (partially) characterize the regime-switching point. In particular, I show that for $\sigma_b^2(N) > 4N\Delta^2$, no algorithm can achieve finite regret in $\mathcal{E}(\Delta)$.

3.1 Minmax Lower Bounds

This section presents minimax lower bounds of order $O(\sqrt{N})$ for the signaled bandit problem with a constant that depends on the relative informational content of signals to rewards. Specifically, lower bounds are decreasing in the information content of signals, i.e. they are increasing in $\sigma_b^2(N)$. This result justifies referring to the experts' problem as the easiest signaled bandit instance, and the bandit

problem as the hardest. In fact, the attained lower bounds interpolate smoothly between known bounds in the experts' and signaled bandit games. In the K-arm problem, the lower bounds $L(\cdot)$ for the experts' and bandit problem known are given by[5]

$$L(\text{Experts'}) = C\sqrt{N\sigma_a^2}, \qquad L(\text{Bandit}) = C\sqrt{N(K-1)\sigma_a^2}$$

Theorem 3.1. *Lower Bound in the Signaled Bandit Problem. For any policy π,*

$$\max_{\nu \in \mathcal{E}} \mathcal{R}(\pi, \nu, N) \geq \frac{1}{27}\sqrt{\frac{N(K-1)\sigma_a^2\sigma_b^2(N)}{\sigma_b^2(N) + (K-2)\sigma_a^2}}$$

Corollary 3.2. *The experts' problem is the easiest signaled bandit game and the bandit problem is the hardest signal bandit game.*

Proof of Corollary 3.2.

a) **Experts' Problem:** For $\sigma_a^2 = \sigma_b^2(N)$, $L(\text{Signaled Bandit}) = C\sqrt{N\sigma_a^2} = L(\text{Experts})$.

b) **Bandit Problem:** For $\sigma_a^2 < \sigma_b^2(N) = \infty$, $L(\text{Signaled Bandit}) = C\sqrt{N(K-1)\sigma_a^2} = L(\text{Bandit})$.

c) **Signaled Bandit Problem:** For $\sigma_a^2 \leq \sigma_b^2(N) \leq \infty$ $L(\text{Signaled Bandit}) \in [L(\text{Experts}), L(\text{Bandit})]$

K Asymptotics. For any fixed N, as the number of arms goes to ∞, the lower bounds depend only on $\sigma_b^2(N)$. This is to be expected because, for any finite σ_b^2, the informational content of an infinite number of signals will always offset the informational content of a single reward. Lower bounds reflect this principle.

3.2 Instance Dependent Lower Bounds

Consider a signaled bandit where the informational content of rewards is N times that of signals, i.e. $\sigma_b^2(N) = \sigma_a^2 N$, so the learner would need N signals about arm k to obtain the informational equivalent of a single reward from arm k. In this extreme setting, it seems obvious that the learner would need to actively play arm k if she wants to learn anything about its mean before the end of the game. In practice, this signaled bandit environment would be indistinguishable from a regular bandit, and so the learner cannot hope for sub-logarithmic regret. Conversely, when signals are nearly as informative as rewards, i.e. $\sigma_b^2(N)/\sigma_a^2 = O(1)$, the learner can achieve finite regret through pure-exploitation algorithms. The intermediate regime, however, is worth characterizing: what happens to regret when $\lim_{N\to\infty} \frac{\sigma_b^2(N)}{\sigma_a^2} = \infty$, yet $\frac{\sigma_b^2(N)}{N} = o_p(1)$? Theorem 3.4 presents the main result in this section by characterizing (an upper bound on) the switching point from finite to asymptotic regret as a function of σ_b^2.

Policy Class Restriction. As it is customary, instance-dependent bounds are only available for a sub-class of *reasonable* policies. A standard practice in bandit settings is to restrict the class of policies to algorithms that achieve at most logarithmic regret in the bandit game (Degenne et al., 2018; Lattimore and Szepesvári, 2020). There is weakly more information in the signaled bandit,

[5] See Lattimore and Szepesvári (2020) or Cesa-Bianchi and Lugosi (2006).

compared to the bandit regime, thus the assumption is even weaker than the one usually imposed in the literature.

Assumption 3.3. *Logarithmic Policies in the Signaled Bandit Game. A policy $\pi \in \Pi_{\log}$ iff for all $N \in \mathbb{N}_+$ exist finite constants C_0, C_1 such that,*

$$\max_{\nu \in \mathcal{E}(\Delta)} \mathcal{R}(\pi, \nu, N) \leq C_0 \Delta + C_1 \sigma_a^2 \frac{\ln(N)}{\Delta}$$

Theorem 3.4. *Instance-Dependent Lower Bounds. For all $\pi \in \Pi_{\log}$,*

$$\max_{\nu \in \mathcal{E}(\Delta)} \mathcal{R}(\pi, \nu, N) \geq \begin{cases} \frac{\sigma_a^2 \sigma_b^2(N)}{2\Delta(\sigma_b^2(N) - \sigma_a^2)} \left(h\left(1 \vee \frac{\sigma_b^2(N)}{4\Delta^2}\right) - \frac{1}{2} \right)_+ & \text{for } \frac{2\Delta^2 N}{h(N)} \leq \sigma_b^2(N) \leq 4N\Delta^2 \\ \frac{\sigma_a^2 \sigma_b^2(N)}{2\Delta(\sigma_b^2(N) - \sigma_a^2)} \left(h(N) - \frac{1}{2} \right)_+ & \text{for } \sigma_b^2(N) > 4N\Delta^2 \end{cases}$$

where $h(N) \sim O(\ln N)$.

Comment on Theorem 3.4. The particular definition of $h(\cdot)$, as well as the proof of Theorem 3.4 can be found in Appendix B. As anticipated, the Theorem shows that no algorithm can achieve finite regret when the informational content of signals is lower than $N\Delta^2$. Although this threshold may seem very high, note that for worst-case Δ—usually of $O(1/\sqrt{N})$—the threshold becomes rather binding. Importantly, I note that my derivation mimics that of Degenne et al. (2018).[6] In fact, the derivation of instance-dependent lower bounds in the signaled bandit setting faces a similar challenge than the one identified in their setting: standard lower bounds become empty in the presence of a *linear* flow of action-independent information. Building on their results, I overcome this problem by relying on unconventional lower-bounds as derived by Garivier et al. (2019). Finally, I should mention that my method is tailored to the near bandit regime, as Theorem 3.4 does deliver empty bounds for highly informative signals $\sigma_b^2(N) \approx \sigma_a^2$.[7]

4 Upper Bounds in the Signaled Bandit

The signaled bandit framework interpolates between the experts' and bandit regimes, and so one may conjecture that an algorithm which interpolates between the optimal algorithms in the each of these regimes could be good candidate to optimal algorithm in the general game. I show this intuition to be exactly correct.

4.1 Optimal Algorithms in Limiting Regimes

In the experts' game there is no trade-off between exploration and exploitation because information is independent of the learner's actions. As a result, pure-exploitation algorithms turn out to be optimal.

Definition 4.1. *Pure-Exploitation Class. A policy π belongs to the pure-exploitation class $\Pi_{exploit}$ iff the policy samples every arm n times (independently of N) and then selects k_i uniformly at random from $\arg\max_k \hat{\mu}_{ki}^{PE}$ for all $i \in [nK+1, N]$, where $\hat{\mu}_{ki}^{PE} = \frac{1}{i} \sum_{s=1}^{i} \omega_s v_{k_s,k,s}$ for some convex weights ω_s.*

[6] In Degenne et al. (2018), the authors derive lower bounds for a similar game to the signaled bandit where the learner observes rewards of unplayed arms with some small probability

[7] Importantly, the lower bounds do not explode as $\sigma_b^2 \to \sigma_a^2$. For small values of σ_b^2, the term in parenthesis becomes zero and so it does the entire bound.

In words, after a fixed-length *warm-up* exploration, pure-exploitation algorithms simply play the arm with the highest empirical mean in every period. I refer to the pure-exploitation policy with no warm-up as Follow-the-Leader (`FtL`). In the bandit regime, the learner only observes the arm of the selected arm so a critical trade-off between exploration and exploitation emerges. Upper Confidence Bound algorithms solve this trade-off through an additive bound that incentives the exploration of arms with a small number of observations. These bounds B_{ki} are selected such that $\sum_{i=1}^{N} \mathbb{P}(\mid \hat{\mu}_{ki} - \mu \mid \leq B_{ki})$ is conveniently small (sumable).

Definition 4.2. *Upper Confidence Bound Class. A policy π belongs to the upper confidence bound class Π_{UCB} iff it samples every arm once and then selects k_i uniformly at random from $\arg\max_k \hat{\mu}^B_{ki} + B_{ki}$ for all $i \in [nK+1, N]$, where $\hat{\mu}^B_{ki} = \frac{1}{N_k(i)} \sum_{s=1}^{i} \mathbb{1}(k_s = k) v_{k_s,k_s,s}$ and $N_k(i) = \sum_{s=1}^{i} \mathbb{1}(k_s = k)$*

I refer to the policy $\pi \in \Pi_{\texttt{UCB}}$ with $B_{ki} = B^{\texttt{UCB}}_{ki} = \sqrt{8\sigma_a^2 \ln(i)/N_k(i)}$ as `UCB`, where $N_k(i) = \sum_{s=1}^{i} \mathbb{1}(k_s = k)$ is the number of times arm k has been played by period i.[8] Lemma 4.3 summarizes standard results on instance-dependent and worst-case upper bounds for `FtL` and `UCB`.

Lemma 4.3. *Upper Bounds on Limiting Regimes.*

$$\mathcal{R}(\textit{FtL}, \nu, N) \leq \frac{16(K-1)\sigma_a^2}{\Delta} \;\; \forall \;\; \nu \in \mathcal{E}_{\exp}(\Delta), \qquad \mathcal{R}(\textit{UCB}, \nu, N) \leq 4(K-1) + \frac{32\ln(N)(K-1)\sigma_a^2}{\Delta} \;\; \forall \;\; \nu \in \mathcal{E}_{bandit}(\Delta)$$

$$\mathcal{R}(\textit{FtL}, \nu, N) \leq 8\sqrt{N(K-1)\sigma_a^2} \;\; \forall \;\; \nu \in \mathcal{E}_{\exp}, \qquad \mathcal{R}(\textit{UCB}, \nu, N) \leq 4(K-1) + 8\sqrt{2N\ln(N)(K-1)\sigma_a^2} \;\; \forall \;\; \nu \in \mathcal{E}_{bandit}$$

Comment on Upper Bounds. For completeness, a proof of Lemma 4.3 can be found in Appendix C. The upper bounds on these algorithms match the intuitions and lower bounds in the previous Section. `FtL` achieves finite instance-dependent regret and $O(\sqrt{N})$ worst-case regret in the experts' game. Similarly, `UCB` obtains logarithmic instance-dependent regret and $O(\sqrt{N \ln N})$ in the bandit game. As a result, both algorithms are optimal up to finite constants in their respective regimes (with the exception of `UCB` which is optimal up to logarithmic constants in the worst-case setting). Once again, the sharp gap between finite and asymptotic instance-dependent regret shows up when comparing the optimal algorithms of the bandit and the experts' regimes.

Shared Structure. Both algorithms are remarkably similar. Every period they both select the arm with the highest index, where the index is just a sum of an empirical average and an exploration term. The size of the exploration term is linked to the need for exploration in the particular regime. In particular, the experts' regime requires no dedicated exploration, and so the exploration term entering the index in `FtL` is zero. The shared structure between the optimal algorithms sets the foundation for a candidate to (near) optimal algorithm in the general game: such algorithm will select the arm with the highest index in every period, where indexes are given by the sum of their empirical average—obtained by efficient combination of signals and rewards—and an exploration term which decreases to 0 as signals become more informative.

[8] There exist more efficient algorithms in the upper confidence class compared to this particular instantiation. See for instance `UCB-MOSS` in Lattimore and Szepesvári (2020). Nonetheless, the gain in efficiency comes at the expense of more complicated analyses and, frequently, larger algorithmic instability. The signaled bandit adaptation of `UCB` in this manuscript is fairly general, thus, I conjecture, would go through taking as reference any other algorithm in the upper confidence bound class.

4.2 A near-optimal Algorithm in the Signaled Bandit Game

Inverse Variance Weighted Sample Average. Given a set of independent observations with different information content, a reasonable estimator for the expectation of an arm is the inverse variance weighted empirical average, i.e.

$$\begin{aligned}\hat{\mu}_{ki}^{\text{IVW}} &= \frac{\frac{N_k(i)}{\sigma_a^2}\frac{1}{N_k(i)}\sum_{s=1}^{i}\mathbb{1}(k_s=k)\nu_{k_s,k_s,s} + \frac{i-N_k(i)}{\sigma_b^2(N)}\frac{1}{i-N_k(i)}\sum_{s=1}^{i}\mathbb{1}(k_s\neq k)\nu_{k_s,k,s}}{\frac{N_k(i)}{\sigma_a^2}+\frac{i-N_k}{\sigma_b^2(N)}} \\ &= \frac{\sigma_b^2(N)\sum_{s=1}^{i}\mathbb{1}(k_s=k)\nu_{k_s,k_s,s} + \sigma_a^2\sum_{s=1}^{i}\mathbb{1}(k_s\neq k)\nu_{k_s,k,s}}{i\sigma_a^2 + N_k(i)(\sigma_b^2(N)-\sigma_a^2)}\end{aligned}$$

The IVW estimator has many desirable properties. In particular, under Gauss-Markov, it is the most efficient linear estimator of the population mean. Technically, $\hat{\mu}_{ki}^{\text{IVW}}$ is not an IVW empirical average as the σ constants are sub-gaussian constants and not variances, which are unknown to the learner. However, any σ-subgaussian random variable has variance weakly less than σ^2, so the $\hat{\mu}_{ki}^{\text{IVW}}$ estimator is a reasonable proxy for the true IVW estimator.[9] $\hat{\mu}_{ki}^{\text{IVW}}$ assigns larger weights to rewards but actively features signals to construct the empirical average, especially in settings with informative signals or settings with many arms.

`FtsL` is Clearly Suboptimal in the Signaled Bandit Game. Before looking into more sophisticated algorithms, one may wonder if the IVW equivalent of pure-exploitation algorithms is (near) optimal in the general signaled bandit game. At the end of the day, excluding the pure bandit regime, the learner does receive non-trivial free side-information which may completely erode the need for dedicated exploration. Let Follow the Signaled Leader (`FtSL`) be the signaled bandit equivalent of `FtL`, i.e. an algorithm which selects the arm with the highest IVW empirical mean $\hat{\mu}_{ki}^{\text{IVW}}$ in every period. Based on the extensive discussion in the lower bound section, it is no surprise that `FtSL` performs poorly in the general signaled bandit regime. For finite $\sigma^2(N) < M$ for all N, the `FtSL` learner may expect finite regret, but when $\sigma_b^2(N) \sim O(N)$, pure-exploitation algorithms cannot attain uniform sublinear regret. Lemma 4.4 formalizes this intuition and shows a very strong result on this regard. When $\sigma_b^2(N) \sim O(N^\alpha)$, the regret of pure-exploitation policies is lower bounded by a function of the same order as $\sigma_b^2(N)$.

Lemma 4.4. *Lower Bound on Pure-Exploitation Algorithms in the Signaled Bandit Game. Let $\nu \in \mathcal{E}(\Delta)$ be a two-arm bandit with $\nu_1 = (N(\mu_1, \sigma_a^2), N(\mu_2, \sigma_b^2(N)))$, $\nu_2 = (N(\mu_1, \sigma_b^2(N)), N(\mu_2, \sigma_a^2))$ and $\Delta = \mu_1 - \mu_2 > 0$. Let policy $\pi \in \Pi_{exploit}$, then*

$$\mathcal{R}(\nu, \pi, N) \geq C_1 \min\left\{N\Delta, \frac{\sigma_a^2 + \sigma_b^2(N)}{\Delta}\right\}$$

In particular, when $\sigma_b^2(N) \geq C_2\Delta f(N)$, $\mathcal{R}(\nu, \pi, N) \geq C_3 f(N)$ for some finite constants C_1, C_2, C_3.

Interpretation of Lemma 4.4. Taking a very simple environment as illustration, Lemma 4.4 argues that for finite signal informativeness σ_b^2, pure-exploitation algorithms may achieve finite regret in the general signaled bandit game.[10] Nevertheless, as the informational content of signals vanishes with N, standard regret guarantees get compromised. Two immediate corollaries follow

[9] Moreover, when the σ-subgaussian random variable is normally distributed, the random variable has variance σ^2. The weighted averages $\hat{\mu}_{ki}^{\text{IVW}}$ are asymptotically normal, therefore the analogy is asymptotically exact.

[10] Appendix C proves a finite upper bound on regret for `FtSL` for finite $\sigma_b^2(N)$.

Corollary 4.5. *Comparative Performance of* `FtSL` *in Signaled Bandit Games.*

1. `UCB` *dominates* `FtSL` *for all* $\nu \in \mathcal{E}$ *such that* $\sigma_b^2(N) \gtrsim O(\sigma_a^2 \ln N)$,

2. `FtSL` *incurs linear regret for all* $\nu \in \mathcal{E}$ *such that* $\sigma_b^2(N) \gtrsim O(\Delta^2 N)$.

Lemma 4.4 and Corollary 4.5 confirm that `FtSL` is clearly suboptimal in low-information regimes and disproofs the belief that pure-exploitation algorithms can *survive* in signaled bandit settings with minimal information guarantees. Dedicated exploration seems necessary in the general signaled bandit class. In particular, an algorithm which completely disregards information coming from signals as `UCB` may worst-case dominate pure-exploitation algorithms that use the additional information like `FtSL`.

`SUCB`: A Near-Optimal Algorithm. Given the necessity of dedicated exploration in the general signaled bandit, I derive high-probability bounds for the IVW estimator. To do so, I leverage the linear structure of $\hat{\mu}_{ki}^{\text{IVW}}$ and show that $\hat{\mu}_{ki}^{\text{IVW}} - \mu$ is $\left(\frac{\sigma_a^2 \sigma_b^2(N)}{i\sigma_a^2 + N_k(i)(\sigma_b^2(N) - \sigma_a^2)}\right)^{-1/2}$-subgaussian (see Lemma C.3). Accordingly, its high-probability bounds are given by

$$B_{ki}^{\text{IVW}} = \sqrt{\frac{8 \ln(i) \sigma_a^2 \sigma_b^2(N)}{i\sigma_a^2 + N_k(i)(\sigma_b^2(N) - \sigma_a^2)}}$$

Define the Signaled Upper Confidence Bound algorithm (`SUCB`) as

Algorithm 1: `SUCB`

Input $K, \sigma_a^2, \sigma_b^2(N), N$
for $i \in 1 : K$, **sample** $k_i = i$
for $i \in K + 1 : N$, **sample** $k_i = \arg\max_k \hat{\mu}_{ki}^{\text{IVW}} + B_{ki}^{\text{IVW}}$

`SUCB` is the main algorithm in this paper. I use the rest of this section to argue that it has many desirable properties. Firstly, it displays automatic exploration reduction as signals become increasingly informative, and interpolates nicely between `FtL` and `UCB`. Secondly, it is instance-dependent optimal, achieving finite regret for mildly informative signals—unlike `UCB`—and logarithmic regret uniformly—unlike `FtSL`. It is also minmax near-optimal up to logarithmic terms.

Definition 4.6. *Effective Exploration Bonus. Let* $B_i(\pi) \geq 0$ *be the* K*-set of exploration bounds of an additive index policy* π*. Define the effective exploration bonus of policy* π *in period* i *as the maximum pair-wise difference across individual exploration bounds, i.e.*

$$B_i(\pi) = \max_{k,k' \in [K]} \mid B_{ki}(\pi) - B_{k'i}(\pi) \mid$$

Lemma 4.7. *Automatic Exploration Reduction.*

$$B_i(\textit{SUCB}) \leq O\left((\sigma_b^2(N) - \sigma_a^2)\sqrt{\ln(i)/i}\right) \quad wp \;\; 1$$

where $B_i(\textit{SUCB}) = (B_{ki}^{IVW})_k$ *and the probability is taken uniformly over all possible arm counts* $N_k(i)$.

Lemma 4.7 highlights that, effectively, the bounds of `SUCB` go to 0 as signals become increas-

ingly informative. More precisely, the maximum pair-wise gap between bounds goes to 0 linearly in $(\sigma_b^2 - \sigma_a^2)$, what reduces exploration automatically as signals become informative. In addition, it is easy to see that $\hat{\mu}_{ki}^{\text{IVW}} \to \hat{\mu}_{ki}^{\text{PE}}$ as $\sigma_b^2(N) \to \sigma_a^2$ uniformly in N. It follows that SUCB effectively converges to FtL as signals become informative. Importantly, SUCB converges to FtL there where FtL is optimal: the experts' regime. A similar analysis holds as $\sigma_b^2(N) \to \infty$. In this case, $B_{ki}^{\text{IVW}} \to B_{ki}^{\text{UCB}}$, hence the effective exploration of SUCB converges to that of UCB. Furthermore, $\hat{\mu}_{ki}^{\text{IVW}} \to \hat{\mu}_{ki}^{\text{B}}$, so SUCB converges to UCB there where UCB is optimal: the bandit regime.

Theorem 4.8. *Upper Bounds of SUCB in the Signaled Bandit Game.*

$$\mathcal{R}(\textit{SUCB}, \nu, N) \leq \begin{cases} (K-1)\left(4 + \min\left\{\frac{32\sigma_a^2\sigma_b^2 \ln\left(\frac{32\sigma_b^2}{\Delta^2}\right)}{\Delta(\sigma_b^2-\sigma_a^2)}, \hat{C}\frac{32\sigma_b^2}{\Delta}\ln\frac{32\sigma_b^2}{\Delta^2}\right\}\right) & \text{if } \sigma_b^2 \leq \frac{N\Delta^2}{32} \\ (K-1)\left(4 + \min\left\{\frac{32\sigma_a^2\sigma_b^2 \ln(N)}{\Delta(\sigma_b^2-\sigma_a^2)}, \hat{C}\frac{32\sigma_b^2}{\Delta}\ln\frac{32\sigma_b^2}{\Delta^2}\right\}\right) & \text{if } \sigma_b^2 > \frac{N\Delta^2}{32}, \end{cases} \quad \forall \nu \in \mathcal{E}(\Delta)$$

Moreover,

$$\mathcal{R}(\textit{SUCB}, \nu, N) \leq 4(K-1) + \min\left\{8\sqrt{\frac{2N\ln(N)(K-1)\sigma_a^2\sigma_b^2(N)}{\sigma_b^2(N)-\sigma_a^2}}, O\left(\sqrt{N\ln(N)(K-1)\sigma_b^2(N)}\right)\right\} \quad \forall \nu \in \mathcal{E}$$

Theorem 4.8 summarizes the main result in this Section. The definition of $\hat{C}$ can be found in Appendix C. The theorem shows that SUCB achieves logarithmic instance-dependent regret uniformly in the signaled bandit game, unlike pure-exploitation algorithms (see Lemma 4.4). Yet, additionally, it preserves some of the nice properties of pure-exploitation algorithms. In particular, it achieves finite instance-dependent regret for mildly informative signals, an impossible task for regular UCB algorithms. Additionally, observe that the transition point from finite to asymptotic regret identified in the Theorem is of the same order $N\Delta^2$ than the transition point in the Lower Bound of Theorem 3.4. This suggest that the lower bounds are order-tight and that SUCB is optimal in a strong sense in the signaled bandit game.[11] The second term ensures that the bound remains meaningful as $\sigma_b^2 \to \sigma_a^2$.

These results emphasize that the learner needs not to trade between finite regret in informative scenarios (FtL) with uniform logarithmic guarantees (UCB). Instead, she can obtain robust guarantees through the automatic exploration reduction featured in SUCB. Finally, in terms of worst-case regret, the algorithm remains minmax near-optimal up to a logarithmic factor.

5 Simulations

Figure 1 plots the evolution of cumulative regret as a function of N for different σ_b^2 values. Similarly, Figure 2 plots cumulative regret in period $N = 2000$ as a function of σ_b^2. There are several take-aways in this simulation evidence, all in accordance to the analytical results in the paper. Starting from the performance of the pure-exploitation algorithms FtL and FtSL, we can see that they obtain what appears to be finite regret in the pure expert's regime (see Lemma C.4). Yet, as $\sigma_b^2 \to \infty$ they cannot guarantee sublinear regret (see Lemma 4.4 and Corollary 4.5.ii), and significantly underperform compared to exploring algorithms (Corollary 4.5.i). This simple simulation evidence disregards the common misconception that for any finite signal informativeness σ_b^2, pure-exploitation algorithms

[11] There exists a window of length $O(1)$ given by $[N\Delta^2/32, 4\Delta^2 N]$ where SUCB only remains instance-dependent optimal up to a logarithmic factor.

recover finite regret. Additionally, it is worth mentioning that the IVW-powered pure-exploitation algorithm `FtSL` significantly outperforms the standard `FtL`. Secondly, textbook `UCB` guarantees logarithmic regret across all σ_b^2 domains and outperforms `FtSL` for $\sigma_b^2 \gtrsim C\sigma_a^2 \ln N$ (Lemma 4.5). Yet, for small values of σ_b^2 it remains overly conservative.

Finally, observe that `SUCB` achieves finite regret for small values of σ_b^2 while preserving logarithmic regret uniformly over σ_b^2 (Theorem 4.8). Additionally, in light of the simulation evidence, I conjecture that (i) `SUCB` uniformly outperforms textbook `UCB`, and (ii) that `SUCB` may attain regret below the minimum regret across the interpolating policies, i.e. there are real gains to convexity beyond robustness (see $\sigma_b^2 \in [100, 500]$ in Figure 2).

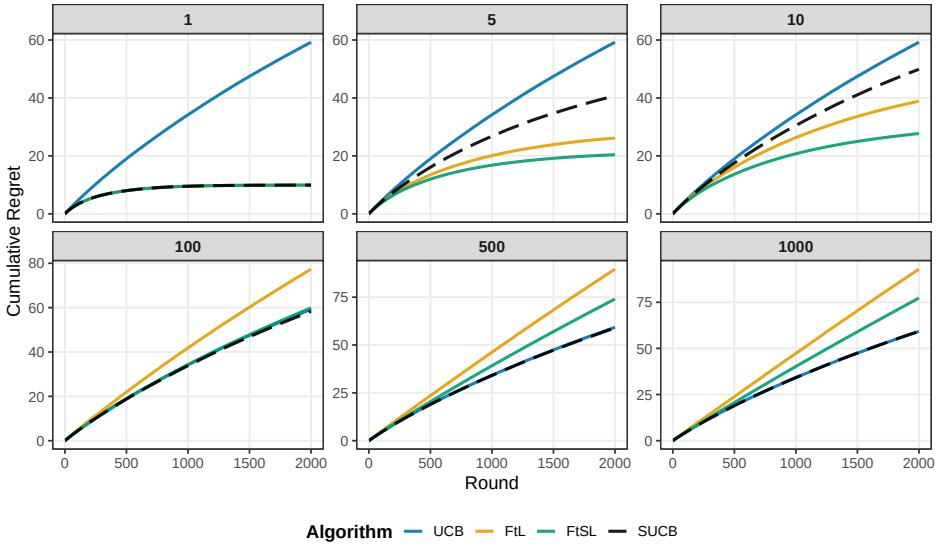


Figure 1: Cumulative Regret across σ_b^2 values (averaged over 10,000 simulations). $\mu = (0, 0.1)$, $\sigma_a^2 = 1$.

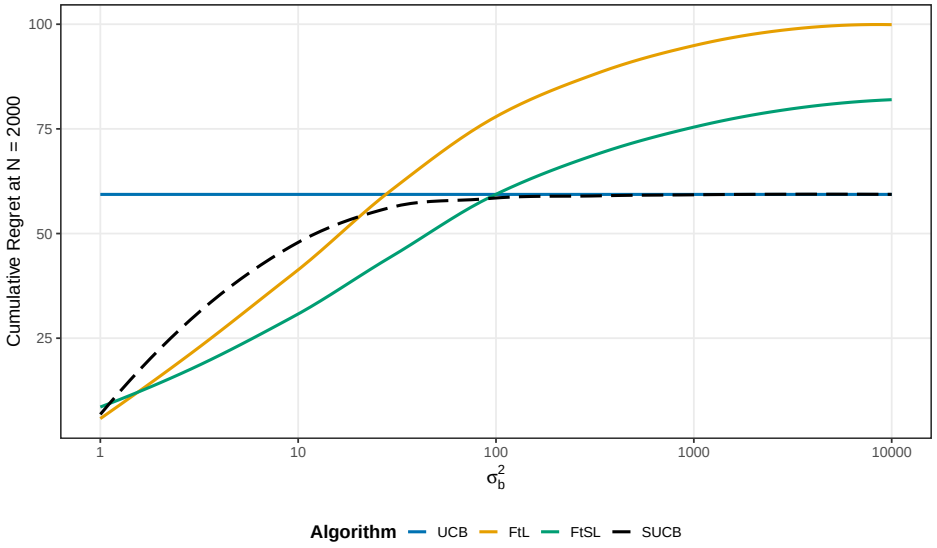


Figure 2: Cumulative Regret at period $N = 2000$ across σ_b^2 values (averaged over 1,000 simulations). $\mu = (0, 0.1)$, $\sigma_a^2 = 1$.

6 Conclusion

This paper introduces the signaled bandit game, an online learning setting which interpolates between experts' and bandit problems through the notion of signal informativeness, parameterized by a subgaussian constant $\sigma^2(N)$. I argued that signal informativeness is a natural interpolating regime that extends the canonical bandit setting in the presence of a free flow of (low quality) information with applications to finance, industrial organization, treatment effects, fraud detection and more. In terms of the difficulty of the game, I first derived worst-case lower bounds of order $\sqrt{N}$ that interpolate smoothly between well known experts' and bandit bounds. More importantly, by allowing informativeness to scale with the horizon (similarly to local-to-zero asymptotics), I was able to derive instance-dependent bounds which feature a sharp transition between finite and logarithmic regret. I identified the transition phase at a point no larger than $CN\Delta^2$.

In terms of algorithmic performance, I showed that pure-exploitation algorithms inherit regret of the same order as the informational content of signals, so they cannot guarantee sublinear regret across the entire signaled bandit spectrum. I then proposed a new algorithm `SUCB`, an inverse variance weighted UCB algorithm with adjusted bounds, and showed that it achieves logarithmic instance-dependent regret and $\sqrt{N \ln N}$ worst case regret in the generalized game, therefore outperforming pure-exploitation algorithms for relatively uninformative signals. More importantly, I showed that `SUCB` achieves finite instance-dependent regret for signals with informational content $\lesssim CN\Delta^2$, i.e. of the same order than the transition phase identified in the lower bounds. As a result, I argued that `SUCB` is order-tight in a strong sense.

Taken together, these results deliver an important message for practitioners: even low-quality information about counterfactual arms can sharply reduce the need for dedicated exploration, hence, auxiliary (unbiased) data should never be discarded. Nevertheless, exploration can still play a decisive role when the informational content of *learning by doing* is high enough. Overall, dedicated exploration yields robust guarantees at the expense of a logarithmic factor of regret. I hope that the signaled bandit may offer a tractable and empirically grounded framework for many decision problems in Economics and Finance where costly experimentation coexists with a steady flow of free, imperfect information.

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Appendix

Before diving into the derivation of the results in this manuscript, I collect without proof a set of results and definitions that will be used repeatedly in my analyses. As a notation clarification, I will write $\sigma_b^2(N)$ as σ_b^2 unless the asymptotic properties of σ_b^2 are relevant for the analysis.

A Preliminary Results

Lemma A.1. *Regret Decomposition - Lemma 4.5 in Lattimore and Szepesvári (2020). For any policy π, environment $\nu \in \mathcal{E}$ and finite horizon N,*

$$\mathcal{R}(\pi, \nu, N) = \sum_{k \neq 1}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)]\Delta_k$$

Lemma A.2. *Linear Combinations of Subgaussian Random Variables - Lemma 5.4 and Corollary 5.5. in Lattimore and Szepesvári (2020). Let $\mathbb{E}[X_i] = \mu$. Asume $X_i - \mu$ are independent, σ_i-subgaussian random variables. Then,*

1. *$c(X_i - \mu)$ is σ-subgaussian for all $c \in \mathbb{R}$,*
2. *$(X_i - \mu) + (X_j - \mu)$ is $\sqrt{\sigma_i^2 + \sigma_j^2}$-subgaussian,*

Moreover, for $\sigma_i^2 = \sigma^2$ for all i, $\hat{\mu} - \mu = \sum_{i=1}^{N}(X_i - \mu)/\sigma$ is $\sigma/\sqrt{N}$-subgaussian.

Lemma A.3. *Divergence Decomposition - Lemma 15.1 in Lattimore and Szepesvári (2020). Consider two environments $\nu, \nu' \in \mathcal{E}$ and some policy π. Let $\mathbb{P}_{kk'}$ be the probability distribution of $\nu_{kk'}$, $\mathbb{P}_k$ be the probability distribution over the vector ν_k, $\mathbb{P}_\nu$ be the probability distribution over the environment matrix ν, and $\mathbb{P}_{\nu,\pi}$ be the probability measure induced by the N-round interaction of policy π and environment ν. $\mathbb{P}'_{kk'}$, $\mathbb{P}'_k$, $\mathbb{P}'$ and $\mathbb{P}_{\nu',\pi}$ are defined as expected. Finally, write $D(\cdot,\cdot)$ for the KL-divergence between two probability measures. Then,*

$$D(\mathbb{P}_{\nu,\pi}, \mathbb{P}_{\nu',\pi}) = \sum_{k=1}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)] \cdot D(\mathbb{P}_k, \mathbb{P}'_k)$$

B Proofs of Section 3

B.1 Preliminary Lemma in Proof of Theorem 3.1

I start by presenting a small extension to Lemma A.3

Lemma B.1. *Divergence Decomposition in Signaled Bandits. Given two signaled bandits $\nu, \nu' \in \mathcal{E}$, and some policy π,*

$$D(\mathbb{P}_{\nu,\pi}, \mathbb{P}_{\nu',\pi}) = \sum_{k=1}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)] \left(\sum_{k'=1}^{K} D(\mathbb{P}_{kk'}, \mathbb{P}'_{kk'}) \right) \tag{2}$$

Proof of Lemma B.1. Remember that $\mathbb{P}_k = (\mathbb{P}_{k1}, \mathbb{P}_{k2}, \ldots, \mathbb{P}_{kK})$. Moreover, under inde-

pendence (and the properties of logarithms) $D(\mathbb{P}_k, \mathbb{P}'_k) = \sum_{k'=1}^{K} D(\mathbb{P}_{kk'}, \mathbb{P}'_{kk'})$. Then,

$$D(\mathbb{P}_{\nu,\pi}, \mathbb{P}_{\nu',\pi}) = \sum_{k=1}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)] \cdot D(\mathbb{P}_k, \mathbb{P}'_k) = \sum_{k=1}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)] \ \left(\sum_{k'=1}^{K} D(\mathbb{P}_{kk'}, \mathbb{P}'_{kk'}) \right)$$

where the first equality follows from Lemma A.3. □

B.2 Proof of Theorem 3.1.

The proof proceeds constructively by showing that at least one of the following two signaled bandit environments will make any policy π incur in non-trivial worst-case regret. Let ν be such that

- $\nu_{11} = N(\Delta, \sigma_a^2)$, $\nu_{k1} = N(\Delta, \sigma_b^2)$ for all $k \neq 1$,
- $\nu_{kk} = N(0, \sigma_a^2)$ for all $k \neq 1$, $\nu_{kk'} = N(0, \sigma_b^2)$ for all $k' \neq k$

Then, for some policy π and fixed horizon N, let arm l be the non-optimal vector arm with the lowest expected number of plays after N rounds, i.e.

$$l = \arg\min_{k \neq 1} \mathbb{E}_{\nu,\pi}[N_k(N)]$$

By definition of l, it follows that $\mathbb{E}_{\nu,\pi}[N_l(N)] \leq \frac{N}{K-1}$. Secondly, let ν' be such that

- $\nu'_{11} = N(\Delta, \sigma_a^2)$, $\nu'_{k1} = N(\Delta, \sigma_b^2)$ for all $k \neq 1, l$,
- $\nu'_{ll} = N(2\Delta, \sigma_a^2)$, $\nu'_{kl} = N(2\Delta, \sigma_b^2)$ for all $k \neq l$,
- $\nu_{kk} = N(0, \sigma_a^2)$ for all $k \neq 1, l$, $\nu_{kk'} = N(0, \sigma_b^2)$ for all $k' \neq k$

In other words, environment ν mimics environment ν' in every arm except for the least sampled arm ν_l. While arm 1 is optimal in ν, arm l is optimal in ν'. It follows that

$$\begin{aligned}
\mathcal{R}(\pi, \nu, N) + \mathcal{R}(\pi, \nu', N) &> \mathbb{P}_{\nu,\pi}(N_1(N) \leq N/2) \cdot \Delta \frac{N}{2} + \mathbb{P}_{\nu',\pi}(N_1(N) > N/2) \cdot \Delta \frac{N}{2} \\
&> \Delta \frac{N}{4} \exp(-D(\mathbb{P}_{\nu,\pi}, \mathbb{P}_{\nu',\pi})) \\
&= \Delta \frac{N}{4} \exp\left(- \sum_{k=1}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)] \ \left(\sum_{k'=1}^{K} D(\mathbb{P}_{kk'}, \mathbb{P}'_{kk'}) \right) \right) \\
&= \Delta \frac{N}{4} \exp\left(- \sum_{k=1}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)] \ D(\mathbb{P}_{kl}, \mathbb{P}'_{kl}) \right) \\
&= \Delta \frac{N}{4} \exp\left(- \mathbb{E}_{\nu,\pi}[N_l(N)] D(\mathbb{P}_{ll}, \mathbb{P}'_{ll}) - \sum_{k \neq l}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)] \ D(\mathbb{P}_{kl}, \mathbb{P}'_{kl}) \right) \\
&= \Delta \frac{N}{4} \exp\left(- \mathbb{E}_{\nu,\pi}[N_l(N)] \frac{2\Delta^2}{\sigma_a^2} - \sum_{k \neq l}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)] \frac{2\Delta^2}{\sigma_b^2} \right) \\
&= \Delta \frac{N}{4} \exp\left(- \mathbb{E}_{\nu,\pi}[N_l(N)] 2\Delta^2 \frac{\sigma_b^2 - \sigma_a^2}{\sigma_a^2 \sigma_b^2} - \sum_{k=1}^{K} \mathbb{E}_{\nu,\pi}[N_k(N)] \frac{2\Delta^2}{\sigma_b^2} \right)
\end{aligned}$$

$$
\begin{aligned}
&= \Delta\frac{N}{4}\exp\left(-\mathbb{E}_{\nu,\pi}[N_l(N)]2\Delta^2\frac{\sigma_b^2-\sigma_a^2}{\sigma_a^2\sigma_b^2} - N\frac{2\Delta^2}{\sigma_b^2}\right)\\
&=\geq \Delta\frac{N}{4}\exp\left(-\frac{2N\Delta^2}{K-1}\frac{\sigma_b^2-\sigma_a^2}{\sigma_a^2\sigma_b^2} - \frac{2N\Delta^2}{\sigma_b^2}\right)\\
&= \Delta\frac{N}{4}\exp\left(-\frac{2N\Delta^2}{K-1}\frac{\sigma_b^2+(K-2)\sigma_a^2}{\sigma_a^2\sigma_b^2}\right)
\end{aligned}
$$

where the second inequality follows from the Bretagnole-Huber inequality, the third equality follows from Lemma B.1, the fourth equality by the selection of environments, the sixth equality by the KL-divergence of normal distributions with equal variances, and the seventh inequality by definition of arm l under policy π. Finally, selecting $\Delta = \sqrt{\frac{(K-1)\sigma_a^2\sigma_b^2}{4N\left(\sigma_b^2+(K-2)\sigma_a^2\right)}}$ leads to

$$
\begin{aligned}
\max_{\nu\in\mathcal{E}}\mathcal{R}(\pi,\nu,N) &\geq \max\{\mathcal{R}(\pi,\nu,N),\mathcal{R}(\pi,\nu',N)\} \geq \frac{\mathcal{R}(\pi,\nu,N)+\mathcal{R}(\pi,\nu',N)}{2}\\
&> \frac{\exp(-1/2)}{16}\sqrt{\frac{N(K-1)\sigma_a^2\sigma_b^2}{\sigma_b^2+(K-2)\sigma_a^2}} > \frac{1}{27}\sqrt{\frac{N(K-1)\sigma_a^2\sigma_b^2}{\sigma_b^2+(K-2)\sigma_a^2}} \qquad \square
\end{aligned}
$$

B.3 Proof of Theorem 3.4

Fix two environments ν,ν' as defined in Subsection B.2. Then, starting from Lemma B.1, for every π

$$
\begin{aligned}
D(\mathbb{P}_{\nu,\pi},\mathbb{P}_{\nu',\pi}) &= \sum_{k=1}^{K}\mathbb{E}_{\nu,\pi}[N_k(N)]\left(\sum_{k'=1}^{K}D(\mathbb{P}_{kk'},\mathbb{P}'_{kk'})\right)\\
&= \mathbb{E}_{\nu,\pi}[N_l(N)]2\Delta^2\frac{\sigma_b^2-\sigma_a^2}{\sigma_a^2\sigma_b^2} + \frac{2\Delta^2}{\sigma_b^2}N \implies\\
\mathbb{E}_{\nu,\pi}[N_l(N)] &= D(\mathbb{P}_{\nu,\pi},\mathbb{P}_{\nu',\pi})\frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)} - \frac{\sigma_a^2}{\sigma_b^2-\sigma_a^2}N
\end{aligned}
$$

I now invoke Equation 6 from Garivier et al. (2019). In this work, the authors show that for any environment pair ν,ν', history-adapted policy π and history-measurable random variable $Z\in[0,1]$

$$
D(\mathbb{P}_{\nu,\pi},\mathbb{P}_{\nu',\pi}) \geq D_{\text{bin}}(\mathbb{E}_{\nu,\pi}[Z],\mathbb{E}_{\nu',\pi}[Z])
$$

where $D_{\text{bin}}(p,q)$ is the KL-divergence for bernouilli random variables with probability p and q, respectively. Clearly, $N_1(N)/N\in[0,1]$ is history measurable, therefore

$$
\begin{aligned}
D(\mathbb{P}_{\nu,\pi},\mathbb{P}_{\nu',\pi}) &\geq D_{\text{bin}}(\mathbb{E}_{\nu,\pi}[N_1/N],\mathbb{E}_{\nu',\pi}[N_1/N])\\
&\geq \frac{\mathbb{E}_{\nu,\pi}[N_1(N)]}{N}\cdot\ln\left(\frac{1}{\frac{\mathbb{E}_{\nu',\pi}[N_1]}{N}}\right) - \ln(2)
\end{aligned}
$$

where the second inequality follows from $D_{\text{bin}}(p,q)\geq p\ln(1/q)-\ln(2)$. Now, applying Lemma A.1 and Assumption 3.3

$$
\begin{aligned}
\mathcal{R}(\pi,\nu,N) &= \sum_{k\neq 1}\mathbb{E}_{\nu,\pi}[N_k(N)]\Delta = \Delta(N-\mathbb{E}_{\nu,\pi}[N_1(N)]) \implies\\
\frac{\mathbb{E}_{\nu,\pi}[N_1(N)]}{N} &= 1-\frac{\mathcal{R}(\pi,\nu,N)}{N\Delta} \geq 1-\frac{1}{N}\left(C_0+C_1\sigma_a^2\frac{\ln(N)}{\Delta^2}\right)
\end{aligned}
$$

and, similarly,

$$\mathcal{R}(\pi,\nu',N) = \mathbb{E}_{\nu',\pi}[N_1(N)]\Delta + \sum_{k\neq 1,l}\mathbb{E}_{\nu',\pi}[N_k(N)]2\Delta \geq \mathbb{E}_{\nu',\pi}[N_1(N)]\Delta \implies$$
$$\frac{\mathbb{E}_{\nu',\pi}[N_1(N)]}{N} \leq \frac{\mathcal{R}(\pi,\nu',N)}{N\Delta} \leq \frac{1}{N}\left(C_0 + C_1\sigma_a^2\frac{\ln(N)}{\Delta^2}\right)$$

Finally, bringing all results together

$$\begin{aligned}
\mathbb{E}_{\nu,\pi}[N_l(N)] &= D(\mathbb{P}_{\nu,\pi},\mathbb{P}_{\nu',\pi})\frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)} - \frac{\sigma_a^2}{\sigma_b^2-\sigma_a^2}N \\
&\geq \frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}\left(\frac{\mathbb{E}_{\nu,\pi}[N_1(N)]}{N}\cdot\ln\left(\frac{1}{\frac{\mathbb{E}_{\nu',\pi}[N_1]}{N}}\right) - \ln(2)\right) - \frac{\sigma_a^2}{\sigma_b^2-\sigma_a^2}N \\
&\geq \frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}\left(\left(1-\frac{1}{N}\left(C_0+C_1\sigma_a^2\frac{\ln(N)}{\Delta^2}\right)\right)\cdot\ln\left(\frac{1}{\frac{1}{N}\left(C_0+C_1\sigma_a^2\frac{\ln(N)}{\Delta^2}\right)}\right) - \ln(2)\right) - \frac{\sigma_a^2}{\sigma_b^2-\sigma_a^2}N \\
&= \frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}h(N) - \frac{\sigma_a^2}{\sigma_b^2-\sigma_a^2}N = \frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}\left(h(N)-\frac{2\Delta^2}{\sigma_b^2}N\right)_+
\end{aligned}$$

where the second-to-last equality serves as definition of $h(N)$, and the last equality emphasizes that $N_l(N)$ is non-negative, therefore the bound is only meaningful for $\sigma_b^2 \geq \frac{2\Delta^2 N}{h(N)}$.

Properties of $h(N)$. I first note that $h(N) \sim O(\ln N)$, and is increasing for $N \geq e$. To show the latter property, I take derivatives of the first a second terms entering the product and show that they are both strictly positive for $N \geq \exp\left(1-\frac{C_0\Delta^2}{C_1\sigma_a^2}\right)$. Some additional algebra shows that for small C_2

$$h'(N) \geq \frac{1}{N}\left(1-\frac{1}{\ln(N)+C_2}\right)$$

such that, for $N \geq e^2$, $h'(N) \geq 1/2N$. For instance, the inequality above holds for $C_2 = \frac{C_0\Delta^2(\sigma_b^2-\sigma_a^2)}{2C_1\sigma_a^2\sigma_b^2}$. It then follows that $\frac{\partial\mathbb{E}_{\nu,\pi}[N_l(N)]}{\partial N} \geq \frac{\sigma_a^2\sigma_b^2}{4N\Delta^2(\sigma_b^2-\sigma_a^2)} - \frac{\sigma_a^2}{\sigma_b^2-\sigma_a^2}$, so $\mathbb{E}_{\nu,\pi}[N_l(N)]$ is increasing (at least) in $\left[e^2, \frac{\sigma_b^2}{4\Delta^2}\right]$. Finally, I note that $\mathbb{E}_{\nu,\pi}[N_l(N)]$ is non-decreasing in N, thus,

$$\mathbb{E}_{\nu,\pi}[N_l(N)] \geq \sup_{3\leq n\leq N}\left(\frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}\left(h(n)-\frac{2\Delta^2}{\sigma_b^2}n\right)_+\right)$$

where 3 has been chosen as the lower bound to ensure that $h(n)$ is increasing. I now evaluate separately two regimes: small N and large N. For small N, $e^2 \leq N \leq \frac{\sigma_b^2}{4\Delta^2}$, from the monotonicity of $\mathbb{E}_{\nu,\pi}[N_l(N)]$, the RHS is increasing in N, therefore,

$$\mathbb{E}_{\nu,\pi}[N_l(N)] \geq \frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}\left(h(N)-\frac{2\Delta^2}{\sigma_b^2}N\right)_+ \geq \frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}\left(h(N)-\frac{1}{2}\right)_+$$

For large N, $N \geq \max\left\{1, \frac{\sigma_b^2}{4\Delta^2}\right\}$, the lower function can be lower bounded by its value at $n = \max\left\{1, \frac{\sigma_b^2}{4\Delta^2}\right\}$, thus

$$\mathbb{E}_{\nu,\pi}[N_l(N)] \geq \frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}\left(h\left(1\vee\frac{\sigma_b^2}{4\Delta^2}\right)-\frac{1}{2}\right)_+$$

Finally,

$$\mathcal{R}(\pi,\nu,N) \geq \mathbb{E}_{\nu,\pi}[N_l(N)]\Delta \geq \begin{cases} \frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}\left(h\left(1\vee\frac{\sigma_b^2}{4\Delta^2}\right)-\frac{1}{2}\right)_+ & \text{for } \frac{2\Delta^2 N}{h(N)} \leq \sigma_b^2 \leq 4N\Delta^2 \\ \frac{\sigma_a^2\sigma_b^2}{2\Delta^2(\sigma_b^2-\sigma_a^2)}\left(h\left(N\right)-\frac{1}{2}\right)_+ & \text{for } \sigma_b^2 > 4N\Delta^2 \end{cases} \qquad \square$$

C Proofs of Section 4

C.1 Proof of Lemma 4.4

When the learner plays arm 1, she observes (the equivalent to) a random reward $Z_i^1 = \nu_{11i} - \nu_{12i}$. Similarly, when playing arm 2, she observes (the equivalent to) a random reward $Z_i^2 = \nu_{21i} - \nu_{22i}$. Crucially, note that $Z_i^1 \sim Z_i^2 \sim N(\Delta, \sigma_a^2 + \sigma_b^2)$. Therefore, after i periods $\hat{\Delta}_i = \frac{1}{i}\sum_{s=1}^{i} Z_s \sim N\left(\Delta, \frac{\sigma_a^2+\sigma_b^2}{i}\right)$. Now, observe that after round nK, the learner selects arm 2 iff $\hat{\Delta}_i < 0$. Therefore, the probability of selecting arm 2 in period i is

$$\mathbb{P}(\hat{\Delta}_i \leq 0) = \Phi\left(-\Delta\sqrt{\frac{i}{\sigma_a^2+\sigma_b^2}}\right)$$

Now, let $m_N = \frac{\sigma_a^2+\sigma_b^2}{\Delta^2}$. Then, observe that $\Phi(-\sqrt{i/m_N})$ is decreasing in i. Therefore $\Phi(-\sqrt{N/m_N}) \leq \Phi(-\sqrt{i/m_N})$ for all $i \leq N$. It follows that

$$\mathbb{E}[N_2(N)] = n + \sum_{i=nK+1}^{N} \Phi\left(-\Delta\sqrt{\frac{i}{\sigma_a^2+\sigma_b^2}}\right) = n + \sum_{i=nK+1}^{N} \Phi\left(-\sqrt{i/m_N}\right) \geq c_0 \min\{N-nK, m_N\}$$

where in the last equality I have used the following result: when $m_N \geq N$, then $\Phi(\cdot) \geq \Phi(-1) = c_0$ for all i, therefore $\sum_{i=nK+1}^{N} \Phi\left(-\sqrt{i/m_N}\right) \geq c_0(N-nK)$. When $m_N < N$ the result above holds for all $nK+1 \leq i \leq m_N$, i.e.

$$\begin{aligned}\sum_{i=nK+1}^{N} \Phi\left(-\sqrt{i/m_N}\right) &= \sum_{i=nK+1}^{m_N} \Phi\left(-\sqrt{i/m_N}\right) + \sum_{i=m_N}^{N} \Phi\left(-\sqrt{i/m_N}\right) \\ &\geq c_0(m_N - (nK+1)) + \sum_{i=m_N}^{N} \Phi\left(-\sqrt{i/m_N}\right) \geq c_0(m_N-(nK+1))\end{aligned}$$

Finally, from Lemma A.1,

$$\mathcal{R}(\nu,\pi,N) = \mathbb{E}[N_2(N)]\Delta \geq c_0 \min\left\{N\Delta, \frac{\sigma_a^2+\sigma_b^2}{\Delta}\right\}$$

The second part of the lemma follows immediately. $\square$

C.2 Proof of Lemma 4.3

For completeness, I collect brief proofs for the results in Lemma 4.3. These results are part of textbook material in Online and Bandit Learning. The main results for the signaled bandit game will mimic the structure and tools laid down in this Subsection, therefore I consider useful their inclusion as part of this manuscript. All the expectations in this Section are taken wrt to the joint measure $\nu \times \pi$, so we write $\mathbb{E}$ for $\mathbb{E}_{\nu,\pi}$ and $\mathbb{P}$ for $\mathbb{P}_{\nu,\pi}$ whenever possible to avoid clutter.

C.3 Upper Bounds on FtL in the Experts' Problem

Note that $\frac{\mu_1+\mu_k}{2} = \mu_k + \frac{\Delta_k}{2} = \mu_1 - \frac{\Delta_k}{2}$. Therefore

$$\begin{aligned}\mathbb{E}[N_k(N)] &\leq \mathbb{E}\left[\sum_{i=1}^{N} \mathbb{1}(\hat{\mu}_{ki} \geq \hat{\mu}_{1i})\right] = \sum_{i=1}^{N} \mathbb{P}\big(\hat{\mu}_{ki} \geq \hat{\mu}_{1i}\big) \\ &\leq \sum_{i=1}^{N} \left(\mathbb{P}\left(\hat{\mu}_{ki} \geq \mu_k + \frac{\Delta_k}{2}\right) + \mathbb{P}\left(\hat{\mu}_{1i} \leq \mu_1 - \frac{\Delta_k}{2}\right)\right) \leq 2\sum_{i=1}^{N} \exp\left(-\frac{i\Delta_k^2}{8\sigma_a^2}\right) \leq \frac{16\sigma_a^2}{\Delta_k^2}\end{aligned}$$

where the main inequality follows from Lemma A.2, and the last step follows from (i) bounding the geometric sum $\sum_{i=1}^{N} \exp\left(-\frac{\Delta_k^2}{8\sigma_a^2}\right)^i$ by an infinite sum, (ii) solving the infinite sum, and (iii) noting that $\exp(-a)/(1-\exp(-a)) \leq 1/a$. Using Lemma A.1, it now follows that for all $\nu \in \mathcal{E}_{\exp}(\Delta)$

$$\mathcal{R}(\texttt{FtL}, \nu, N) = \sum_{k\neq 1}^{K} \mathbb{E}[N_k(N)]\Delta_k \leq \sum_{k\neq 1}^{K} \mathbb{E}[N_k(N)]\Delta_k \leq \sum_{k\neq 1}^{K} \frac{16\sigma_a^2}{\Delta_k} \leq \frac{16(K-1)\sigma_a^2}{\Delta}$$

Moreover, for all $\nu \in \mathcal{E}_{\exp}$

$$\begin{aligned}\mathcal{R}(\texttt{FtL}, \nu, N) &\leq \sum_{k\neq 1, \Delta_k < \Delta}^{K} \mathbb{E}[N_k(N)]\Delta_k + \sum_{k\neq 1, \Delta_k \geq \Delta}^{K} \mathbb{E}[N_k(N)]\Delta_k \\ &\leq N\Delta + \sum_{k\neq 1, \Delta_k \geq \Delta}^{K} \frac{16\sigma_a^2}{\Delta_k} \leq N\Delta + \frac{16\sigma_a^2(K-1)}{\Delta} \leq 8\sqrt{N(K-1)\sigma_a^2}\end{aligned}$$

where the last equality follows from setting $\Delta = \sqrt{16(K-1)\sigma_a^2/N}$. □

C.4 Upper Bounds on UCB in the Bandit Problem

I first introduce some preliminary lemmas.

Lemma C.1. *Conditions for UCB. Let $B_{ki} = B_{ki}^{UCB}$ be the bound of arm k in period i. By definition of UCB, every time arm $k \neq 1$ is sampled in period $i+1$, at least one of the following conditions must be true*

$$(1)\ \ \hat{\mu}_{1i} + B_{1i} \leq \mu_1 \qquad (2)\ \ \hat{\mu}_{ki} - B_{ki} > \mu_k \qquad (3)\ \ 2B_{ki} > \Delta_k \tag{3}$$

Proof of Lemma C.1. Let (2) and (3) be false, then

$$\hat{\mu}_{1i} + B_{1i} \overset{\texttt{UCB}}{\leq} \hat{\mu}_{ki} + B_{ki} = \hat{\mu}_{ki} - B_{ki} + 2B_{ki} \overset{(2)}{\leq} \mu_k + 2B_{ki} \overset{(3)}{\leq} \mu_k + \Delta_k = \mu_1$$

Let (1) and (3) be false, then

$$\hat{\mu}_{ki} - B_{ki} = \hat{\mu}_{ki} + B_{ki} - 2B_{ki} \overset{\texttt{UCB}}{\geq} \hat{\mu}_{1i} + B_{1i} - 2B_{ki} \overset{(1)}{\geq} \mu_1 - 2B_{ki} \overset{(3)}{\geq} \mu_1 - \Delta_k = \mu_k$$

Let (1) and (2) be false, then

$$\Delta_k = \mu_1 - \mu_k \overset{(1),(2)}{<} \hat{\mu}_{1i} + B_{1i} - (\hat{\mu}_{ki} - B_{ki}) \overset{\texttt{UCB}}{\leq} \hat{\mu}_{ki} + B_{ki} - (\hat{\mu}_{ki} - B_{ki}) = 2B_{ki}$$

Lemma C.2. *Probability of* `UCB` *Conditions.*

$$(3) \text{ is TRUE} \iff N_k(i) < \frac{32\sigma_a^2 \ln(i)}{\Delta_k^2}, \qquad \mathbb{P}(\ (1)\ \text{is TRUE} \mid (3)) \le i^{-3}, \qquad \mathbb{P}(\ (2)\ \text{is TRUE} \mid (3)) \le i^{-3}$$

Proof of Lemma C.2. Condition (3) is immediate given the definition of $B_{ki}^{\texttt{UCB}}$. Conditions (1) and (2) follow from Lemma A.2 and a standard *skipping argument*

$$\mathbb{P}(\hat{\mu}_{1i} + B_{1i} \le \mu_1 \mid (3)) \le \sum_{n=1}^{i} \mathbb{P}(N_k(i) = n)\mathbb{P}(\hat{\mu}_1(n) + B_1(n) \le \mu_1) \le \sum_{n=1}^{i} i^{-4} \le i^{-3}$$

where I note that Condition (3) avoids the pathological $n = 0$ and that $\hat{\mu}_k$ is $\sigma_a/\sqrt{N_k}$-subgaussian. Condition (2) is completely symmetric. □

Remark of Lemma C.2. Set $\tilde{i}_k = \left\{i : N_k(i) \ge \frac{32\sigma_a^2 \ln(N)}{\Delta_k^2}\right\}$, then, for all $i \ge \tilde{i}_k$, Condition (3) in Lemma C.2 is violated.

Proof of Lemma 4.3. For all $\nu \in \mathcal{E}_{\text{bandit}}(\Delta)$

$$\begin{aligned}
\mathcal{R}(\texttt{UCB}, \nu, N) &= \sum_{k\neq 1}^{K} \mathbb{E}[N_k(N)]\Delta_k = \sum_{k\neq 1}^{K}\sum_{i=1}^{N} \mathbb{P}(k_i = k)\Delta_k \\
&\le \sum_{k\neq 1}^{K} \left(\tilde{i}_k\Delta_k + \sum_{i=\tilde{i}_k+1}^{N} \mathbb{P}(k_i = k)\Delta_k \right) \le \sum_{k\neq 1}^{K} \left(\tilde{i}_k\Delta_k + 2\Delta_k \sum_{i=\tilde{i}+1}^{N} i^{-3} \right) \\
&\le \sum_{k\neq 1}^{K} \left(\tilde{i}_k\Delta_k + 4\Delta_k\right) = 4\sum_{k\neq 1}^{K}\Delta_k + \sum_{k\neq 1}^{K} \frac{32\sigma_a^2 \ln(N)}{\Delta_k} \le 4(K-1) + \frac{32(K-1)\sigma_a^2 \ln(N)}{\Delta}
\end{aligned}$$

where the first equality follows from Lemma A.3, the fourth inequality from Lemmas C.1 and C.2, the fifth inequality from noting that $\sum_i^N i^{-3} \le 2$ and the last inequality from noting that $\Delta_k \in [\Delta, 1]$ for all k. A similar analysis holds for all $\nu \in \mathcal{E}_{\text{bandit}}$

$$\begin{aligned}
\mathcal{R}(\texttt{UCB}, \nu, N) &\le N\Delta + \sum_{k\neq 1, \Delta_k \ge \Delta}^{K} \mathbb{E}[N_k(N)]\Delta_k \le N\Delta + 4\sum_{k\neq 1, \Delta_k \ge \Delta}^{K} \Delta_k + \sum_{k\neq 1, \Delta_k \ge \Delta}^{K} \frac{32\sigma_a^2 \ln(N)}{\Delta_k} \\
&\le N\Delta + 4(K-1) + \frac{32(K-1)\sigma_a^2 \ln(N)}{\Delta} \le 4(K-1) + 8\sqrt{2N\ln(N)(K-1)\sigma_a^2}
\end{aligned}$$

where the last inequality follows from setting $\Delta = \sqrt{\frac{24\ln(N)(K-1)\sigma_a^2}{N}}$. □

C.5 Subgaussianity of $\hat{\mu}_{ki}^{\texttt{IVW}}$

Lemma C.3. *Subgaussianity of IVW estimator. Let* $\hat{\mu}_k = \frac{1}{N_k}\sum_{i=1}^{N_k} X_{k,i}$ *where* $X_{k,1}, \ldots, X_{k,N_k}$ *is a sequence of independent* σ_k*-subgaussian random variables. Let*

$$\hat{\mu}_k^{IVW} = \sum_{k=1}^{K} w_k\hat{\mu}_k, \qquad \text{where} \quad w_k = \frac{N_k/\sigma_k^2}{\sum_{k'=1}^{K} N_{k'}/\sigma_{k'}^2}$$

Then, $\hat{\mu}_k^{IVW}$ *is* $\left(\sum_{k=1}^{K} \frac{N_k}{\sigma_k^2}\right)^{-1/2}$*-subgauss. And, for any* $\nu \in \mathcal{E}$, $\hat{\mu}_k^{IVW}$ *is* $\left(\frac{\sigma_a^2\sigma_b^2}{N\sigma_a^2 + N_k(\sigma_b^2 - \sigma_a^2)}\right)^{1/2}$*-subgauss.*

Proof of Lemma C.3. Let $\sum_{k'}^{K} N_k/\sigma_k^2 = V$. From Lemma A.2 it follows that (i) $w_k\hat{\mu}_k$ is $w_k\sigma_k/\sqrt{N_k}$-subgaussian and (ii) $\hat{\mu}_k^{\mathrm{IVW}}$ is $\sqrt{\sum_{k=1}^{K} w_k^2\sigma_k^2/N_k}$-subgaussian. Simplifying,

$$\sum_{k=1}^{K} w_k^2 \frac{\sigma_k^2}{N_k} = \sum_{k=1}^{K} \left(\frac{N_k/\sigma_k^2}{V}\right)^2 = \frac{1}{V^2}\sum_{k=1}^{K}\frac{N_k}{\sigma_k^2} = \frac{1}{V}$$

Therefore, $\hat{\mu}_k^{\mathrm{IVW}}$ is $(\sum_{k'}^{K} N_k/\sigma_k^2)^{-1/2}$-subgaussian. The second part of the lemma follows by replacing σ_k^2 by σ_a^2 and σ_b^2 accordingly. □

C.6 Auxiliary Upper Bound Proof for FtSL in the Signaled Bandit Problem

Lemma C.4. *Upper Bounds on FtSL in the Signaled bandit Game.*

$$\mathcal{R}(\mathit{FtSL}, \nu, N) \le \frac{16\sigma_b^2}{\Delta_k} \le \frac{16(K-1)\sigma_b^2}{\Delta} \quad \forall \nu \in \mathcal{E}(\Delta)$$

Moreover,

$$\mathcal{R}(\mathit{FtSL}, \nu, N) \le 8\sqrt{N(K-1)\sigma_b^2} \quad \forall \nu \in \mathcal{E}$$

Proof of Lemma C.4. Recall $\hat{\mu}_{ki} = \hat{\mu}_{ki}^{\mathrm{IVW}}(N_k(i), \sum_{k'\neq k} N_{k'}(i))$. Then,

$$\begin{aligned}
\mathbb{E}[N_k(N)] &\le \mathbb{E}\left[\sum_{i=1}^{N} \mathbb{1}\big(\hat{\mu}_{ki} > \hat{\mu}_{1i}\big)\right] = \sum_{i=1}^{N} \mathbb{P}\big(\hat{\mu}_{ki} > \hat{\mu}_{1i}\big) \\
&= \sum_{i=1}^{N}\sum_{s=0}^{i}\sum_{t=0}^{i-s} \mathbb{P}(N_k(i)=s, N_1(i)=t)\cdot \mathbb{P}\big(\hat{\mu}_{ki}(s,i-s) > \hat{\mu}_{1i}(t,i-t)\big) \\
&\le \sum_{i=1}^{N}\sum_{s=0}^{i}\sum_{t=0}^{i-s} \mathbb{P}(N_k(i)=s, N_1(i)=t)\cdot \left[\mathbb{P}\left(\hat{\mu}_{ki}(s,i-s) - \mu_k > \frac{\Delta_k}{2}\right) + \mathbb{P}\left(\mu_1 - \hat{\mu}_{1i}(t,i-t) > \frac{\Delta_k}{2}\right)\right] \\
&\le \sum_{i=1}^{N}\sum_{s=0}^{i}\sum_{t=0}^{i-s} \mathbb{P}(N_k(i)=s, N_1(i)=t)\cdot \left[\exp\left(-\frac{\Delta_k^2(s\sigma_b^2 + (i-s)\sigma_a^2)}{8\sigma_a^2\sigma_b^2}\right) + \exp\left(-\frac{\Delta_k^2(t\sigma_b^2 + (i-t)\sigma_a^2)}{8\sigma_a^2\sigma_b^2}\right)\right] \\
&\le \sum_{i=1}^{N}\sum_{s=0}^{i}\sum_{t=0}^{i-s} \mathbb{P}(N_k(i)=s, N_1(i)=t)\cdot \left[\exp\left(-\frac{i\Delta_k^2}{8\sigma_b^2}\right) + \exp\left(-\frac{i\Delta_k^2}{8\sigma_b^2}\right)\right] = \sum_{i=1}^{N} 2\exp\left(-\frac{i\Delta_k^2}{8\sigma_b^2}\right) \le \frac{16\sigma_b^2}{\Delta_k^2}
\end{aligned}$$

where the main inequality follows from Lemma C.3. Now, using Lemma A.1, for all $\nu \in \mathcal{E}(\Delta)$

$$\mathcal{R}(\mathtt{FtSL}, \nu, N) = \sum_{k\neq 1}^{K} \mathbb{E}[N_k(N)]\Delta_k \le \sum_{k\neq 1}^{K} \frac{16\sigma_b^2}{\Delta_k} \le \frac{16(K-1)\sigma_b^2}{\Delta}$$

Moreover, for all $\nu \in \mathcal{E}$

$$\begin{aligned}
\mathcal{R}(\mathtt{FtSL}, \nu, N) &= \sum_{k\neq 1, \Delta_k < \Delta}^{K} \mathbb{E}[N_k(N)]\Delta_k + \sum_{k\neq 1, \Delta_k \ge \Delta}^{K} \mathbb{E}[N_k(N)]\Delta_k \\
&\le N\Delta + \sum_{k\neq 1, \Delta_k \ge \Delta}^{K} \frac{16\sigma_b^2}{\Delta_k} \le N\Delta + \frac{16\sigma_b^2(K-1)}{\Delta} \le 8\sqrt{N(K-1)\sigma_b^2}
\end{aligned}$$

where the last equality follows by setting $\Delta = \sqrt{16(K-1)\sigma_b^2/N}$. □

C.7 Proof of Lemma 4.7.

Note that

$$B_i(\pi) = \max_{k,k' \in [K]} \mid B_{ki}(\pi) - B_{k'i}(\pi) \mid = \max_{k \in [K]} B_{ki}(\pi) - \min_{k' \in [K]} B_{k'i}(\pi)$$

B_{ki} is decreasing in $N_k(i)$, $\arg\max_{k \in [K]} B_{ki}(\pi) = \arg\min_k N_k(i)$ and $\arg\min_{k' \in [K]} B_{k'i}(\pi) = \arg\max_{k'} N_{k'}(i)$. Let $N_{\max} - N_{\min} \leq i$, then,

$$\begin{aligned}
B_i &= \sqrt{\frac{8\ln(i)\sigma_a^2\sigma_b^2}{i\sigma_a^2 + N_{\min}(i)(\sigma_b^2 - \sigma_a^2)}} - \sqrt{\frac{8\ln(i)\sigma_a^2\sigma_b^2}{i\sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)}} \\
&= \sqrt{8\ln(i)\sigma_a^2\sigma_b^2}\frac{\sqrt{i\sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)} - \sqrt{i\sigma_a^2 + N_{\min}(i)(\sigma_b^2 - \sigma_a^2)}}{\sqrt{i\sigma_a^2 + N_{\min}(i)(\sigma_b^2 - \sigma_a^2)}\sqrt{i\sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)}} \\
&= \sqrt{8\ln(i)\sigma_a^2\sigma_b^2}\frac{(\sigma_b^2 - \sigma_a^2)(N_{\max}(i) - N_{\min}(i))}{\sqrt{i\sigma_a^2 + N_{\min}(i)(\sigma_b^2 - \sigma_a^2)}\sqrt{i\sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)}\left(\sqrt{i\sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)} + \sqrt{i\sigma_a^2 + N_{\min}(i)(\sigma_b^2 -}\right.} \\
&\leq \sqrt{8\ln(i)\sigma_a^2\sigma_b^2}\frac{(\sigma_b^2 - \sigma_a^2)(N_{\max} - N_{\min})}{2(i\sigma_a^2)^{3/2}} \\
&\leq (\sigma_b^2 - \sigma_a^2)\frac{\sigma_b}{\sigma_a^2}\sqrt{2\ln(i)/i} \qquad \square
\end{aligned}$$

C.8 Proof of Theorem 4.8

`SUCB` operates on the same logic as `UCB`, therefore Lemma C.1 holds immediately. I now present a small variation to Lemma C.2 for the signaled bandit game.

Lemma C.5. *Probability of `SUCB` Conditions.*

$$(3) \text{ is TRUE} \iff i\sigma_a^2 + N_k(i)(\sigma_b^2 - \sigma_a^2) < \frac{32\sigma_a^2\sigma_b^2\ln(i)}{\Delta_k^2}, \qquad \mathbb{P}(\,(1) \text{ is TRUE} \mid (3)) \leq i^{-3}, \qquad \mathbb{P}(\,(2) \text{ is TRUE} \mid (3)) \leq i^{-}$$

Proof of Lemma C.5. Condition (3) is immediate given the definition of B_{ki}^{IVW}. Conditions (1) and (2) follow from $\left(\frac{i\sigma_a^2 + N_k(i)(\sigma_b^2 - \sigma_a^2)}{\sigma_a^2\sigma_b^2}\right)^{-1/2}$-subgaussianity of $\hat{\mu}_k^{\text{IVW}}$ (see Lemma C.3) and the skipping trick. $\square$

Remark of Lemma C.5. I distinguish two regimes where Condition (3) is violated.

1. Let $i_k^* = \min\left\{\frac{32\sigma_b^2}{\Delta_k^2}, N\right\}$. Observe that for $\sigma_a^2 < \sigma_b^2 < \infty$, $F(i) = \frac{32\sigma_a^2\sigma_b^2\ln(i)}{\Delta_k^2(\sigma_b^2 - \sigma_a^2)} - i\frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2}$ is concave with a unique maximum in i_k^*. Let $\tilde{i}_k = \left\{i : N_k(i) \geq \frac{32\sigma_a^2\sigma_b^2\ln(i_k^*)}{\Delta_k^2(\sigma_b^2 - \sigma_a^2)} - i_k^*\frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2}\right\}$, then, for all $i \geq \tilde{i}_k$, Condition (3) is violated.

2. Let $\hat{i}_k = \frac{32\sigma_b^2}{\Delta_k^2}\left(\ln\left(\frac{32\sigma_b^2}{\Delta_k^2}\right) + \ln\ln\left(\frac{32\sigma_b^2}{\Delta_k^2}\right) + \frac{32\sigma_b^2}{\Delta_k^2}\right) = \hat{C}\frac{32\sigma_b^2}{\Delta_k^2}\ln\frac{32\sigma_b^2}{\Delta_k^2}$, it follows from Lambert's W that for $i \geq \hat{i}_k$, Condition (3) is also violated for arm k.

I now use each of these regimes to derive slightly different upper bounds. Eventually, I combine both of them to obtain a tighter bound.

Proof of Theorem 4.8. (1) $\tilde{i}_k$ regime. For all $\nu \in \mathcal{E}(\Delta)$

$$\begin{aligned}
\mathcal{R}(\texttt{SUCB}, \nu, N) &= \sum_{k\neq 1}^{K} \mathbb{E}[N_k(N)]\Delta_k = \sum_{k\neq 1}^{K}\sum_{i=1}^{N} \mathbb{P}(k_i = k)\Delta_k \\
&\leq \sum_{k\neq 1}^{K} \left(\tilde{i}_k \Delta_k + \sum_{i=\tilde{i}_k+1}^{N} \mathbb{P}(k_i = k)\Delta_k \right) \leq \sum_{k\neq 1}^{K} \left(\tilde{i}_k \Delta_k + 2\Delta_k \sum_{i=\tilde{i}+1}^{N} i^{-3} \right) \\
&\leq \sum_{k\neq 1}^{K} \left(\tilde{i}_k \Delta_k + 4\Delta_k \right) = 4\sum_{k\neq 1}^{K} \Delta_k + \sum_{k\neq 1}^{K} \left(\frac{32\sigma_a^2\sigma_b^2 \ln(i_k^*)}{\Delta_k(\sigma_b^2 - \sigma_a^2)} - i_k^* \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2}\Delta_k \right) \\
&\leq 4(K-1) + \frac{32(K-1)\sigma_a^2\sigma_b^2 \ln\left(\min\left\{ \frac{32\sigma_b^2}{\Delta^2}, N \right\} \right)}{\Delta(\sigma_b^2 - \sigma_a^2)}
\end{aligned}$$

Result follows,

$$\mathcal{R}(\texttt{SUCB}, \nu, N) \leq \begin{cases} 4(K-1) + \frac{32(K-1)\sigma_a^2\sigma_b^2 \ln\left(\frac{32\sigma_b^2}{\Delta^2}\right)}{\Delta(\sigma_b^2 - \sigma_a^2)} & \text{if } \sigma_b^2 \leq \frac{N\Delta^2}{32} \\ 4(K-1) + \frac{32(K-1)\sigma_a^2\sigma_b^2 \ln(N)}{\Delta(\sigma_b^2 - \sigma_a^2)} & \text{if } \sigma_b^2 > \frac{N\Delta^2}{32} \end{cases}$$

Therefore, SUCB achieves finite regret for sufficiently small σ_b^2. In fact, the switching point shares the same order (up to a constant factor) as the switching point identified in the lower bound. (2) $\hat{i}_k$ regime. For all $\nu \in \mathcal{E}(\Delta)$,

$$\begin{aligned}
\mathcal{R}(\texttt{SUCB}, \nu, N) &= \sum_{k\neq 1}^{K} \mathbb{E}[N_k(N)]\Delta_k = \sum_{k\neq 1}^{K}\sum_{i=1}^{N} \mathbb{P}(k_i = k)\Delta_k \\
&= \sum_{k\neq 1} \left(\hat{i}_k \Delta_k + \sum_{i=\hat{i}_k+1}^{N} \mathbb{P}(k_i = k)\Delta_k \right) \leq \hat{C}(K-1)\frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} + \sum_{k\neq 1}^{K} \left(2\Delta_k \sum_{i=\tilde{i}+1}^{N} i^{-3} \right) \\
&\leq \hat{C}(K-1)\frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} + \sum_{k\neq 1}^{K} 4\Delta_k \leq \hat{C}(K-1)\frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} + 4(K-1)
\end{aligned}$$

Combining both results,

$$\mathcal{R}(\texttt{SUCB}, \nu, N) \leq \begin{cases} (K-1)\left(4 + \min\left\{ \frac{32\sigma_a^2\sigma_b^2 \ln\left(\frac{32\sigma_b^2}{\Delta^2}\right)}{\Delta(\sigma_b^2 - \sigma_a^2)}, \hat{C}\frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} \right\} \right) & \text{if } \sigma_b^2 \leq \frac{N\Delta^2}{32} \\ (K-1)\left(4 + \min\left\{ \frac{32\sigma_a^2\sigma_b^2 \ln(N)}{\Delta(\sigma_b^2 - \sigma_a^2)}, \hat{C}\frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} \right\} \right) & \text{if } \sigma_b^2 > \frac{N\Delta^2}{32} \end{cases}$$

Finally, one can extend the analysis for all $\nu \in \mathcal{E}$.

$$\begin{aligned}
\mathcal{R}(\texttt{SUCB}, \nu, N) &\leq N\Delta + \sum_{k\neq 1, \Delta_k \geq \Delta}^{K} \mathbb{E}[N_k(N)]\Delta_k \leq N\Delta + 4\sum_{k\neq 1, \Delta_k \geq \Delta}^{K} \Delta_k + \sum_{k\neq 1, \Delta_k \geq \Delta}^{K} \left(\frac{32\sigma_a^2\sigma_b^2 \ln(i_k^*)}{\Delta_k(\sigma_b^2 - \sigma_a^2)} - i_k^* \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2}\Delta_k \right) \\
&\leq N\Delta + 4(K-1) + \frac{32(K-1)\sigma_a^2\sigma_b^2 \ln\left(\min\left\{ \frac{32\sigma_b^2}{\Delta^2}, N \right\} \right)}{\Delta(\sigma_b^2 - \sigma_a^2)} \\
&\leq N\Delta + 4(K-1) + \frac{32(K-1)\sigma_a^2\sigma_b^2 \ln(N)}{\Delta} \leq 4(K-1) + 8\sqrt{\frac{2N\ln(N)(K-1)\sigma_a^2\sigma_b^2}{\sigma_b^2 - \sigma_a^2}}
\end{aligned}$$

where the last inequality follows from setting $\Delta = \sqrt{\frac{32(K-1)\sigma_a^2\sigma_b^2\ln(N)}{N(\sigma_b^2-\sigma_a^2)}}$. Similarly,

$$\begin{aligned}\mathcal{R}(\texttt{SUCB},\nu,N) &\leq N\Delta + \sum_{k\neq 1,\Delta_k\geq\Delta}^{K}\mathbb{E}[N_k(N)]\Delta_k \leq N\Delta + \hat{C}(K-1)\frac{32\sigma_b^2}{\Delta}\ln\frac{32\sigma_b^2}{\Delta^2} + \sum_{k\neq 1,\Delta_k\geq\Delta}^{K}\sum_{i=\hat{i}}^{N}\mathbb{P}(k_i=k)\Delta_k \\ &\leq N\Delta + \hat{C}(K-1)\frac{32\sigma_b^2}{\Delta}\ln\frac{32\sigma_b^2}{\Delta^2} + 4(K-1) \leq 4(K-1) + O\left(\sqrt{32\hat{C}(K-1)\sigma_b^2\ln\left(\frac{32\sigma_b^2 N}{32\hat{C}(K-1)\sigma_b^2}\right)}\right) \\ &= 4(K-1) + O\left(\sqrt{N\ln(N)(K-1)\sigma_b^2}\right)\end{aligned}$$

where the last equality follows from setting $\Delta = \sqrt{\frac{32\hat{C}(K-1)\sigma_b^2\ln(N)}{N}}$. Combining both results, I obtain the worst-case bound

$$\mathcal{R}(\texttt{SUCB},\nu,N) \leq 4(K-1) + \min\left\{8\sqrt{\frac{2N\ln(N)(K-1)\sigma_a^2\sigma_b^2}{\sigma_b^2-\sigma_a^2}}, O\left(\sqrt{N\ln(N)(K-1)\sigma_b^2}\right)\right\} \qquad \square$$